GeoModels Tutorial: simulation, estimation and prediction of spatio-temporal data using Gaussian random fields

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Introduction

In this tutorial we show how to analyze geo-referenced spatio-temporal data using Gaussian random fields (RFs) with the R package `GeoModels` [Bevilacqua et al., 2026]. We first load the R libraries needed for the analysis and set the name of the model in the `GeoModels` package:

```
rm(list=ls())
require(GeoModels)
require(fields)
model="Gaussian"  # model name in the GeoModels package
set.seed(12)
```

Simulation of a space-time Gaussian random field

Let us consider a space-time Gaussian RF $Z = \{Z(\boldsymbol{s}, t), \boldsymbol{s} \in \mathcal{S}, t \in \mathcal{B}\}$, where $\boldsymbol{s}$ represents a location in the spatial domain $\mathcal{S}$ and t represents a temporal instant in the domain $\mathcal{B}$. We assume that Z is stationary with zero mean, unit variance and correlation function given by $\rho(\boldsymbol{h}, u) = \text{cor}\{Z(\boldsymbol{s}+\boldsymbol{h}, t+u), Z(\boldsymbol{s}, t)\}$.

Then we consider the RF $Y = \{Y(\boldsymbol{s}, t), \boldsymbol{s} \in \mathcal{S}, t \in \mathcal{B}\}$ defined by the location and scale transformation:

$$Y(\boldsymbol{s}, t) = \mu(\boldsymbol{s}, t) + \sigma Z(\boldsymbol{s}, t). \tag{1}$$

where $\mu(\boldsymbol{s}, t) = X(\boldsymbol{s}, t)^\top \boldsymbol{\beta}$ and $X(\boldsymbol{s}, t)$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^\top$ is a k-dimensional vector of unknown parameters (in this tutorial we fix $k = 2$). Then $\text{E}\{Y(\boldsymbol{s}, t)\} = X(\boldsymbol{s}, t)^\top \boldsymbol{\beta}$, $\text{Var}\{Y(\boldsymbol{s}, t)\} = \sigma^2$ and $\text{Cov}\{Y(\boldsymbol{s}+\boldsymbol{h}, t+u), Y(\boldsymbol{s}, t)\} = \sigma^2 \rho(\boldsymbol{h}, u)$.

Suppose we want to simulate a realization of Y at $t_1 = 1, t_2 = 2, \ldots, t_T = 30$, $T = 30$ temporal instants and $N = 400$ spatial locations uniformly distributed in the unit square. The total number of space-time observations is $NT = 12000$. We first set the temporal instants, spatial coordinates and associated covariates.

```
coordt=1:30       # number of temporal instants
T=length(coordt)
NN=400           # number of spatial locations
```

```
x=runif(NN,0,1); y=runif(NN,0,1)
coords=cbind(x,y)
X=cbind(rep(1,NN*T),runif(NN*T))
```

We then specify the mean, variance and nugget parameters.

```
mean=0.5; mean1=-0.25
sill=2; nugget=0
```

where `mean`, `mean1` and `sill` are respectively β_1, β_2 and σ^2.

In this tutorial we assume a simple spatially isotropic, symmetric in time, separable

Wendland model for the space-time correlation function, that is

$$\rho(\boldsymbol{h}, u; \alpha_s, \alpha_t, \mu_s, \mu_t) = \left(1 - \frac{\|\boldsymbol{h}\|}{\alpha_s}\right)_+^{\mu_s} \left(1 - \frac{|u|}{\alpha_t}\right)_+^{\mu_t}, \tag{2}$$

where $x_+ = \max(x, 0)$. This correlation is compactly supported in space and time: $\rho(\boldsymbol{h}, u) = 0$ whenever $\|\boldsymbol{h}\| \geq \alpha_s$ or $|u| \geq \alpha_t$. This feature permits the use of sparse-matrix algorithms for the covariance matrix.

We set the name of the correlation model and the associated parameters:

```
corrmodel="Wend0_Wend0"
scale_s=0.2; scale_t=2
```

where `scale_s` and `scale_t` correspond to the compact-support radii α_s and α_t. We are now ready to simulate, using Cholesky decomposition, the space-time Gaussian random field through the function `GeoSim`:

```
param=list(nugget=0,mean=mean,mean1=mean1,
           scale_t=scale_t,scale_s=scale_s,sill=sill,
           power2_s=4,power2_t=4)
ss1=GeoSim(coordx=coords,coordt=coordt,corrmodel=corrmodel,
           X=X,sparse=TRUE,model=model,param=param)$data
```

The option `sparse=TRUE` allows sparse matrix methods for Cholesky factorization through the package `spam` [Furrer and Sain, 2010]. Information about sparsity is obtained with `GeoCovmatrix`:

```
cc=GeoCovmatrix(coordx=coords,coordt=coordt,
                corrmodel=corrmodel,X=X,sparse=TRUE,
                model=model,param=param)
```

```
cc$nozero
[1] 0.01768
```

This means that approximately 98% of the covariance-matrix entries are zero.

Estimation of a Gaussian space-time random field

Given a realization $\{Y(\boldsymbol{s}_i, t_l),\ l = 1, \ldots, T,\ i = 1, \ldots, N\}$, let $f_Y(y_{il}, y_{jk}; \boldsymbol{\theta})$ be the Gaussian joint density of a pair of observations $Y(\boldsymbol{s}_i, t_l)$ and $Y(\boldsymbol{s}_j, t_k)$. Then the pairwise composite log-likelihood is defined as [Bevilacqua and Gaetan, 2015]

$$pl(\boldsymbol{\theta}) = \sum_{(i,j,l,k)\in\mathcal{D}} \log f_Y(y_{il}, y_{jk}; \boldsymbol{\theta})\, w_{ijlk}, \tag{3}$$

where the set $\mathcal{D}$ counts each distinct pair once:

$$\mathcal{D} = \{(i,j,l,k) :\ 1 \le l \le k \le T, \quad \begin{cases} 1 \le i < j \le N, & l = k, \\ 1 \le i, j \le N, & l < k. \end{cases} \}.$$

The non-negative weights w_{ijlk} do not depend on $\boldsymbol{\theta}$ and are based on spatial nearest neighbors and a time-lag threshold:

$$w_{ijlk}(m, d_t) = \begin{cases} 1, & \boldsymbol{s}_i \in \mathcal{N}_m(\boldsymbol{s}_j), \quad |t_l - t_k| \le d_t, \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

Here $\mathcal{N}_m(\boldsymbol{s}_j)$ is the set of m spatial nearest neighbors of $\boldsymbol{s}_j$. In this case $\boldsymbol{\theta} = (\beta_1, \beta_2, \sigma^2, \alpha_s, \alpha_t)^\top$. The pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{pl}$ is obtained by maximizing (3) with respect to $\boldsymbol{\theta}$.

In the `GeoModels` package, fixed and estimated parameters can be specified. Pairwise likelihood estimation is performed using `GeoFit` and BFGS:

```
start=list(mean=mean,mean1=mean1,scale_s=scale_s,
           scale_t=scale_t,sill=sill)
fixed=list(nugget=nugget,power2_s=4,power2_t=4)
fit=GeoFit(data=ss1,coordx=coords,coordt=coordt,
           corrmodel=corrmodel,neighb=5,maxtime=1,X=X,
           optimizer="BFGS",start=start,fixed=fixed)
```

The options `neighb=5` and `maxtime=1` specify the spatial neighbor number and the maximum temporal lag used in the weights (4).

The fitted object contains the parameter estimates, for example the following *historical output from the original tutorial*:

```
Maximum Composite-Likelihood Fitting of Gaussian Random Fields
Setting: Marginal Composite-Likelihood
Model: Gaussian; Distance: Eucl; Type: Pairwise
Covariance model: Wend0_Wend0
Optimizer: BFGS
Number of spatial coordinates: 400
Number of temporal realisations: 30
Number of estimated parameters: 5
Type of convergence: Successful
Maximum log-Composite-Likelihood:  -610875.85
Estimated parameters:
mean     mean1   scale_s   scale_t     sill
0.4704  -0.2221   0.2060   2.0429   2.0337
```

Checking model assumptions

Given the estimates of regression and variance parameters, the estimated residuals

$$\widehat{Z}(\boldsymbol{s}_i, t_l) = \frac{Y(\boldsymbol{s}_i, t_l) - X(\boldsymbol{s}_i, t_l)^\top \widehat{\boldsymbol{\beta}}}{\sqrt{\widehat{\sigma}^2}}, \quad i = 1, \ldots, N,\ l = 1, \ldots, T,$$

can be viewed approximately as a realization of a stationary Gaussian RF with zero mean, unit variance and correlation $\rho(\boldsymbol{h}, u)$. The residuals are computed using `GeoResiduals`:

```
res=GeoResiduals(fit)
```

The marginal Gaussian assumption can be graphically checked with a QQ-plot (Figure 1, left panel) using `GeoQQ`:

```
GeoQQ(res)
```

The correlation assumption can be checked by comparing empirical and fitted space-time semivariograms using `GeoVariogram` and `GeoCovariogram` (Figure 1, right panels):

```
# Fixed locations: use coordx=coords (not coordx_dyn).
vario=GeoVariogram(data=res$data,coordx=coords,
                   coordt=coordt,maxdist=0.6,maxtime=4)
GeoCovariogram(res,vario=vario,fix.lagt=1,fix.lags=1,
                show.vario=TRUE,pch=20)
```

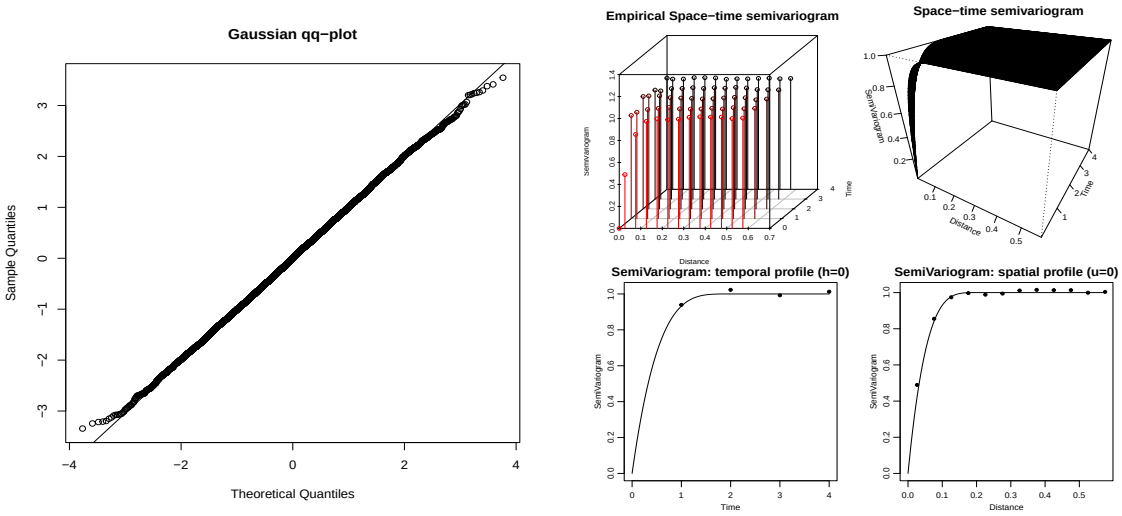


Figure 1: Left: QQ-plot for the residuals of the space-time Gaussian RF. Right: space-time empirical vs. estimated semivariogram function for the residuals. Figure preserved from the original tutorial.

Prediction of space-time Gaussian random fields

For a given space-time location $(\boldsymbol{s}_0, t_0)$ with covariates $X(\boldsymbol{s}_0, t_0)$, the optimal linear predictor of the Gaussian field is computed as

$$\widehat{Y}(\boldsymbol{s}_0, t_0) = X(\boldsymbol{s}_0, t_0)^\top \widehat{\boldsymbol{\beta}} + \sum_{l=1}^{T} \sum_{i=1}^{N} \lambda_{l,i} \left[Y(\boldsymbol{s}_i, t_l) - X(\boldsymbol{s}_i, t_l)^\top \widehat{\boldsymbol{\beta}} \right], \tag{5}$$

where $\boldsymbol{\lambda} = R^{-1}\boldsymbol{c}$, with

- $\boldsymbol{c} = (\text{cor}(Y(\boldsymbol{s}_0, t_0), Y(\boldsymbol{s}_1, t_1)), \ldots, \text{cor}(Y(\boldsymbol{s}_0, t_0), Y(\boldsymbol{s}_N, t_T)))^\top$;
- R is the correlation matrix of all NT observations.

The fitted quantities are treated as known in this plug-in predictor. Kriging is performed using `GeoKrig`. We specify a regular spatial grid and a temporal instant to predict:

```
xx=seq(0,1,0.02)
loc_to_pred=as.matrix(expand.grid(xx,xx))
n_loc=nrow(loc_to_pred)
times=c(2)
n_tim=length(times)
Xloc=cbind(rep(1,n_loc*n_tim),runif(n_loc*n_tim))
```

Then the prediction is calculated from the estimated parameters:

```
pr=GeoKrig(fit,loc=loc_to_pred,time=times,
           Xloc=Xloc,sparse=TRUE,mse=TRUE)
```

A map of observed data at time $t = 2$, kriging predictions, and the associated MSE (Figure 2) is constructed by

```
par(mfrow=c(1,3))
colour=rainbow(100)
quilt.plot(coords[,1],coords[,2],ss1[2,],
           col=colour,main="Time=2")
image.plot(xx,xx,matrix(pr$pred,ncol=length(xx)),
           col=colour,main="Kriging Time=2",ylab="")
image.plot(xx,xx,matrix(pr$mse,ncol=length(xx)),
           col=colour,main="MSE Time=2",ylab="")
```

Local space-time kriging can speed up prediction using a fixed neighborhood; `GeoKrigloc` can be parallelized:

```
prloc=GeoKrigloc(fit,loc=loc_to_pred,time=times,
                 Xloc=Xloc,sparse=TRUE,mse=TRUE,
                 neighb=150,maxtime=2,parallel=TRUE)
summary(c(pr$pred-prloc$pred))
```

Both methods yield similar predictions in the original tutorial, although the historical numerical summary was produced with a grid increment of 0.03 rather than 0.02 (the step in the accompanying R code).

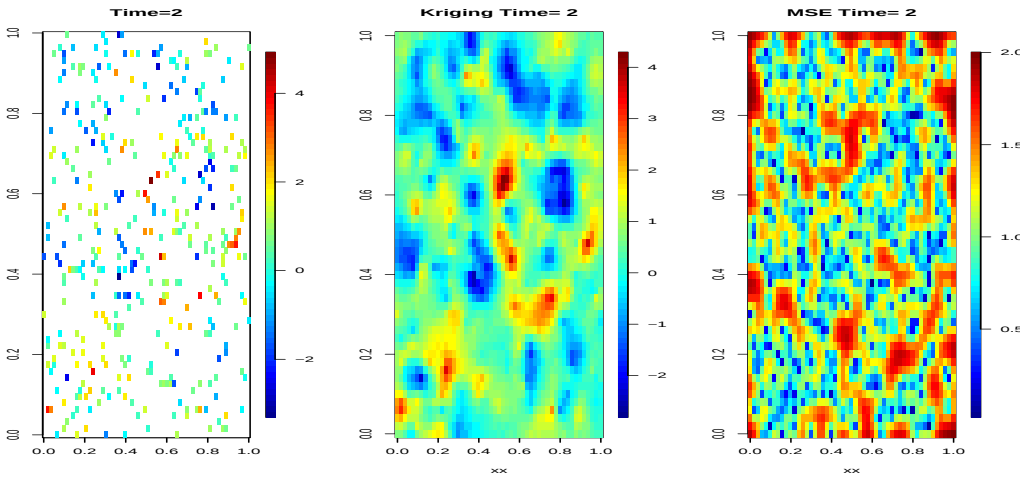


Figure 2: From left to right: observed spatial data at time $t = 2$, associated kriging map and mean square error map. Original figure retained; it was not regenerated with the updated script.

The original tutorial reported the following summary of differences between global and local predictions (with the original 0.03 grid):

```
        Min.      1st Qu.       Median         Mean
-2.636e-04 -1.585e-06  3.590e-09 -2.051e-06
   3rd Qu.         Max.
 2.058e-06  2.931e-04
```

References

Moreno Bevilacqua and Carlo Gaetan. Comparing composite likelihood methods based on pairs for spatial gaussian random fields. *Statistics and Computing*, 25:877–892, 2015. doi: 10.1007/s11222-014-9460-6.

Moreno Bevilacqua, Christian Caamaño-Carrillo, Francisco Cuevas-Pacheco, and Víctor Morales-Oñate. *GeoModels: Procedures for Gaussian and Non Gaussian Geostatistical (Large) Data Analysis*, 2026. URL `https://CRAN.R-project.org/package=GeoModels`. R package version 2.3.0.

Reinhard Furrer and Stephan R. Sain. spam: a sparse matrix R package with emphasis on MCMC methods for gaussian markov random fields. *Journal of Statistical Software*, 36 (10):1–25, 2010. doi: 10.18637/jss.v036.i10.